

GFOR: Grid-forming VSC modelling in RAMSES

Overview. An MMC-type converter connected to the grid through its transformer (resistance R , inductance L , ideal ratio r ; no LC filter). The DC side of the VSC is not modelled (v_{dc} is assumed constant); the focus is on the AC grid dynamics. Per-unit values refer to the nominal apparent power S_{nom} of the converter; the conversion factor of currents is $k = r S_{base}/S_{nom}$.

Reference frame. The d axis is attached to the internal phase angle δ_m of the modulated voltage:

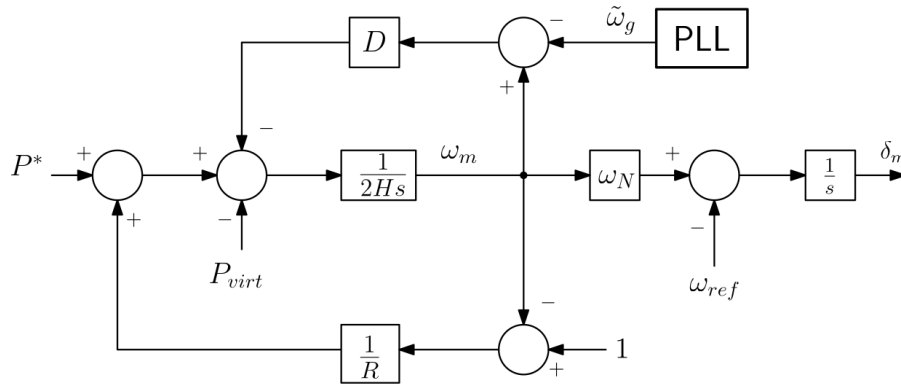
$$v_d = v_x \cos \delta_m + v_y \sin \delta_m, \quad v_q = -v_x \sin \delta_m + v_y \cos \delta_m$$

and similarly for the converter-side currents $i_{xt} = k i_x$, $i_{yt} = k i_y$. The voltage-current relationship in the transformer, in the d - q frame, is:

$$v_{md} = \frac{v_d}{r} + R i_d - \omega_m L i_q, \quad v_{mq} = \frac{v_q}{r} + R i_q + \omega_m L i_d$$

Active power and phase angle control. “Virtual synchronous machine” with inertia emulation (H) and oscillation damping (D):

$$2H \frac{d\omega_m}{dt} = P^* - P_{virt} - D(\omega_m - \tilde{\omega}_g) + \frac{1 - \omega_m}{R_{droop}}, \quad \frac{d\delta_m}{dt} = \omega_N(\omega_m - \omega_{ref})$$



$P_{virt} = (v_d/r) i_d^* + (v_q/r) i_q^*$ is the “virtual power” computed from the unsaturated currents. It amounts to the active power P when the current is not limited, and yields improved large-disturbance stability under limitation of the current. Participation in primary frequency control is optional, with droop R_{droop} (a very large value disables it).

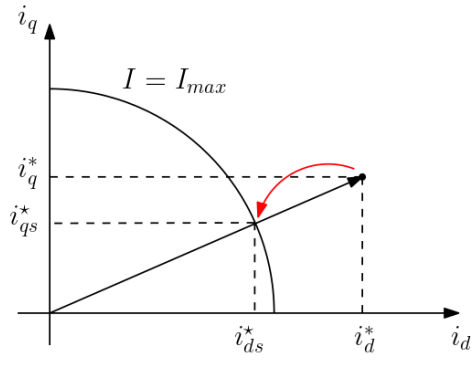
PLL. The purpose of the PLL is only to estimate the grid angular frequency $\tilde{\omega}_g$ for the damping term. Same model as in the grid-following converter (see the GFOL note), with gains $K_{p\omega} = 10/(\omega_N T_{pll})$ and $K_{i\omega} = 25/(\omega_N T_{pll}^2)$, $\omega_N = 2\pi f_N$. The PLL is frozen when the PCC voltage falls below 0.4 pu and reactivated above 0.5 pu (fixed thresholds, with hysteresis).

Current limitation. The currents (i_d^*, i_q^*) that would result from normal voltage control ($v_{md} = V_m^0$, $v_{mq} = 0$) are determined from:

$$V_m^0 = \frac{v_d}{r} + R i_d^* - \omega_m L i_q^*, \quad 0 = \frac{v_q}{r} + R i_q^* + \omega_m L i_d^*$$

The overload ratio $\rho = \sqrt{(i_d^*)^2 + (i_q^*)^2} / I_{max}$ is passed through a first-order lag of time constant T_e (typically 1 ms) with a non-windup lower limit of 1, giving ρ_s . Both components are decreased in the same proportion, and the modulated voltage is set to the value that yields the saturated currents:

$$i_{ds}^* = i_d^* / \rho_s, \quad i_{qs}^* = i_q^* / \rho_s, \quad v_{md} = \frac{v_d}{r} + R i_{ds}^* - \omega_m L i_{qs}^*, \quad v_{mq} = \frac{v_q}{r} + R i_{qs}^* + \omega_m L i_{ds}^*$$



lf $I < I_{max}$: $\rho_s = 1$, $i_d^* = i_{ds}^* = i_d$, $i_q^* = i_{qs}^* = i_q$, $v_{md} = V_m^0$ and $v_{mq} = 0$.

Data. The record has the following structure (default values correspond to a 1100 MVA converter):

INJEC GFOR NAME BUS FP FQ P Q R L r Snom H D Tpl1 Rdroop Imax Te ;

Parameter	Description	Unit	Default
R	resistance of connection transformer	pu	0.005
L	inductance of connection transformer	pu	0.15
r	ratio of ideal transformer		1.02
Snom	nominal apparent power	MVA	1100
H	inertia constant	s	5.0
D	damping constant of virtual synchronous machine		300
Tpl1	time constant of PLL	s	0.1
Rdroop	droop of primary frequency control	pu	999
Imax	maximum current (typically 1–1.2)	pu	1.0
Te	time constant of current saturation	s	0.001

The setpoints V_m^0 (V_{mo}) and P^* (P_o) are computed at initialization from the power-flow solution and can be modified during the simulation with CHGPRM disturbances.

Example:

INJEC GFOR HVDC A 1. 1. 0. 0. 0.005 0.15 1.02 1100. 5.0 300. 0.100 999. 1.0 0.001 ;

Observables: P, Pvirt, Q, Vm, vmd, vmq, deltam, omegam, w_pll, I, id, id*, id*s, iq, iq*, iq*s, rho, rhos.